Design for Manufacturing & Mechanical Part Practice

A practical handbook for translating a 3D CAD model into a manufacturable, inspectable, and assemblable component. Prepared as searchable test data for the Atlas AI RAG module.

Document purpose: Give Atlas AI a manufacturing-aware basis for answering questions about process choice, geometry limits, tolerances, holes, threads, inspection, and design changes.

Document control

Field	Value
Document ID	ATLAS-MFG-DFM-001
Version	1.0
Prepared	7 November 2026
Units	Millimetres (mm) unless stated otherwise
Audience	Mechanical designers, CAD users, manufacturing students, and engineering AI systems
Scope	General design guidance; exact limits depend on supplier, machine, material, tooling, standards, and production volume

How to use this handbook

Treat numerical values in a real project as requirements that must be confirmed with the manufacturing supplier. The guidelines here are qualitative or illustrative unless explicitly stated otherwise. This handbook does not replace ISO/ASME standards, material datasheets, process specifications, or supplier capability statements.

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1. Manufacturing process selection

A CAD model should reflect the process that will create the part. The best process depends on geometry, material, quantity, required properties, surface finish, tolerance, available equipment, lead time, and total cost.

Process	Typical geometry / use	Strengths	Main design concerns
CNC milling	Prismatic blocks, pockets, slots, profiles, complex 3D surfaces	Accurate features, broad material choice, good for low-to-medium volumes	Tool access, internal corner radii, setups, stock removal, thin walls
CNC turning	Rotational parts such as shafts, bushings, pins, pulleys	Efficient for axisymmetric geometry and concentric features	Turning axis, chucking, tool access, cross-holes and secondary operations
Sheet metal	Brackets, covers, panels, enclosures	Efficient for thin, constant-thickness products	Bend radius, bend relief, flat pattern, hole-to-bend spacing, springback
Additive manufacturing	Complex channels, lattices, prototypes, low-volume parts	Geometry freedom and minimal dedicated tooling	Build orientation, supports, anisotropy, surface finish, powder/resin access
Injection moulding	High-volume polymer parts	Low unit cost at scale and repeatability after tooling	Draft, uniform walls, ribs, gates, parting line, undercuts, shrinkage
Casting	Complex metal shapes and larger near-net forms	Can create complex bulk geometry with less machining than a fully machined part	Draft, wall transitions, feeding, porosity risk, machining allowance
Welded fabrication	Frames, brackets, structures made from cut stock or plate	Flexible for low-volume and large structures	Weld access, distortion, joint prep, sequencing, inspection

Process selection questions

- How many parts are required now and over the product life?
- Is the part a prototype, one-off fixture, spare, or production component?
- Which material and mechanical properties are required?
- Which dimensions are functionally critical, and what tolerances can the supplier repeatedly achieve?
- Can the tool reach every surface that must be machined, inspected, or finished?
- Does the selected process create distortion, residual stress, anisotropy, shrinkage, or surface limitations that affect function?
- What secondary processes are required: heat treatment, coating, deburring, machining, or inspection?

DFM principle: Choose the manufacturing process early enough that it can influence the CAD architecture. A part designed for milling may be unnecessarily expensive if it could be formed from sheet metal or assembled from standard sections.

2. CNC machining design rules

CNC machining removes material using cutting tools controlled by a machine program. The CAD model must allow the cutter, holder, workholding, and inspection equipment to access the required geometry.

Milling considerations

- Provide tool access. Deep narrow pockets may require long tools, increasing deflection, chatter, and cycle time.
- Avoid sharp internal corners when possible. A rotating end mill leaves an internal radius; specify a corner radius compatible with an available cutter or use a relief feature where a square internal corner is required.
- Keep wall thickness reasonably uniform and avoid unsupported thin walls that may vibrate or distort during cutting.

- Consider workholding. The part needs a way to be clamped without damaging finished surfaces or blocking tool access.
- Minimise unnecessary setups by grouping features that can be machined from the same orientation.
- Use standard drill sizes, cutter sizes, and thread sizes when function allows; custom sizes may increase tooling and inspection burden.
- Specify surface finish only where function requires it. A blanket fine-finish requirement increases cost.

Turning considerations

- Define a clear axis of rotation and prioritise concentric features that can be produced in one setup.
- Avoid unnecessary changes in diameter and very long slender proportions without checking deflection and support requirements.
- Add lead-in chamfers or radii where they help assembly and tool entry.
- Plan cross-holes, flats, keyways, and non-axisymmetric features as secondary operations if the lathe cannot produce them directly.
- Identify datum surfaces for runout, concentricity, and length inspection when those characteristics matter.

Machining feature guidance

CAD feature	Manufacturing implication	Better design practice
Deep pocket	Long tool reach and possible chatter	Reduce depth, widen pocket, add corner radii, or split the design if function permits.
Sharp internal corner	Cannot be made with a standard round cutter without relief or special process	Specify a practical inside radius or add a tool-clearance relief.
Thin wall	May flex or vibrate during machining	Increase stiffness, support the wall, machine in stages, or use a different process.
Blind drilled hole	Drill point creates a conical bottom; depth is not a flat-bottom depth by default	Specify usable depth and bottom form when function depends on it.
Small deep hole	Tool breakage, chip evacuation, and positional accuracy become difficult	Confirm supplier limits; use appropriate drilling strategy and access.

3. Sheet metal design

Sheet metal parts are commonly made by cutting flat stock and bending it into shape. The CAD model should support both the formed part and its flat pattern.

Core terms

Term	Meaning
Thickness (t)	Nominal sheet thickness. A design often uses one thickness throughout the part.
Inside bend radius	Radius on the inside of a formed bend; depends on material, thickness, tooling, and process.
Bend allowance	Length of material consumed by the bend, used to calculate flat pattern length.
K-factor	Ratio describing the neutral-axis location through sheet thickness for bend calculations. It is process- and material-dependent.
Bend relief	Local cut near a bend to reduce tearing, interference, or material distortion.
Springback	Elastic recovery after bending; the formed angle can differ from the tool angle.

Sheet metal design checklist

- Use a consistent nominal thickness unless a change is functionally necessary and can be formed or joined.
- Choose inside bend radii using material and supplier guidance; do not assume every material can be bent to a very small radius without cracking.

- Keep holes, slots, and cutouts far enough from bend zones to avoid distortion. The required clearance depends on thickness, radius, tooling, and feature size.
- Use bend reliefs at corners where two bends meet or where material could tear or buckle.
- Avoid tiny tabs and narrow flanges that are difficult to form and handle.
- Check the flat pattern for overlaps, self-intersections, and manufacturable blank shape.
- Confirm the bend table or K-factor with the manufacturing supplier; generic bend calculations are estimates until calibrated.

SOLIDWORKS connection: Use the Sheet Metal tools to define base flanges, edge flanges, hems, jogs, and bends. Validate the Flat-Pattern feature and export a DXF/DWG only after confirming the material thickness, bend parameters, and orientation.

4. Additive manufacturing design

Additive manufacturing creates parts layer by layer. Design constraints differ by technology, material, machine, and build orientation. Fused filament fabrication (FFF/FDM), powder bed fusion, stereolithography (SLA), and material jetting have different feature limits and post-processing needs.

General rules

- Choose build orientation based on strength direction, surface finish, support need, and critical dimensions.
- Remember that many processes are anisotropic: strength and stiffness can differ between build-plane and build-direction axes.
- Avoid unsupported overhangs or provide support geometry where the process requires it. The allowable angle is machine- and material-dependent.
- Provide drainage and powder-removal openings for enclosed cavities in processes that use liquid resin or powder.
- Avoid sealed hollow volumes unless the manufacturing process can reliably clear trapped material.
- Allow machining stock on critical surfaces when printed accuracy or finish is insufficient.
- Use lattice or topology-optimised structures only with a validation plan for cleaning, inspection, and mechanical performance.

Mesh export checks

- Choose STL or 3MF settings appropriate to the printer and required surface quality.
- Check that the exported mesh is watertight and correctly oriented.
- Inspect tessellation on circles, fillets, and curved surfaces; coarse mesh settings can alter dimensions.
- Confirm units. A millimetre model interpreted as inches can produce a severe scale error.
- Consider supports, brim/raft, build plate access, and removal strategy before releasing the file.

5. Injection moulding design

Injection moulding forces molten polymer into a closed tool cavity and holds pressure while the material cools. Tooling cost is significant, so the part should be designed for reliable filling, packing, cooling, ejection, and repeatability.

Key design rules

- Aim for reasonably uniform wall thickness. Large abrupt thickness changes can cause sink marks, voids, differential shrinkage, and warpage.
- Use draft on walls parallel to the mould opening direction to help the part release from the tool. Draft amount depends on material, texture, depth, and tool conditions.
- Use ribs to increase stiffness instead of simply making the entire wall much thicker. Rib thickness and root geometry must be confirmed with the moulder to reduce sink and stress concentration.
- Use bosses to support fasteners, but connect them with ribs where needed and avoid overly thick boss bases.
- Plan the parting line, gate location, ejector locations, and slides for undercuts early in design.
- Avoid trapped air and poor flow paths. A moulder may need vents, gates, or flow simulation to confirm filling.
- Account for material shrinkage and warpage through tool compensation and process validation.

Draft, ribs, bosses, and undercuts

Element	Reason	Design question
Draft	Reduces friction and damage during ejection	Which faces are parallel to the opening direction? Is texture present?
Rib	Adds stiffness with less material than a thick wall	Can the rib be thinned or relocated to avoid sink marks on the opposite surface?
Boss	Provides a fastener or locating feature	Can the boss be cored or rib-supported to reduce thick sections?
Undercut	Geometry blocks straight tool opening	Can the geometry be redesigned, or does the tool need a side action/lifter?
Parting line	Separates the two mould halves	Will flash or a witness line affect a critical surface?

Do not universalise moulding numbers: Draft angle, wall thickness, rib ratio, and boss geometry depend on polymer, tool finish, geometry, and supplier capability. Treat common rules of thumb as starting points for a tooling review.

6. Casting and welded fabrication

Casting forms a component by filling a mould with liquid material and allowing it to solidify. Welded fabrication joins cut or formed components using a welding process. Both methods can create shapes that are difficult or expensive to machine from a single billet.

Casting design considerations

- Use draft on faces that must withdraw from the pattern or die.
- Avoid abrupt changes in section thickness; use gradual transitions to reduce hot spots and shrinkage defects.
- Use fillets at internal and external transitions where appropriate to reduce stress concentration and improve material flow.
- Identify likely parting lines, cores, and machining allowances.
- Do not place critical machined dimensions on as-cast surfaces without an explicit tolerance and finishing plan.

- Review porosity risk, feeding, venting, and heat treatment requirements with the casting supplier.

Welded fabrication considerations

- Provide access for the welding torch, filler material, clamps, and inspection.
- Select joint types appropriate to loads, material thickness, and available welding process.
- Consider weld distortion and residual stress. Symmetric joints and a planned welding sequence can reduce distortion.
- Avoid unnecessary weld length and oversized welds; weld size should be based on structural need and applicable standards.
- Provide clearance for weld toes, finishing, coating, and assembly access.
- Specify weld symbols, size, length, location, quality level, and inspection requirements on the drawing according to the applicable standard.

Machining after casting or welding

A cast or welded blank may require machining of datums, bores, sealing faces, or bearing seats. Identify which surfaces are as-formed and which are machined. Add machining allowance only where it is needed and confirm that enough material remains after process variation. Datum selection should reflect how the part will be fixtured and inspected.

7. Holes, threads, tolerances, and fits

Dimensions and tolerances control whether parts can be manufactured and assembled. Tight tolerances increase inspection and production cost, so tolerance should be allocated according to function.

Hole and thread definitions

Feature	Specify	Common mistake
Clearance hole	Fastener standard/size, clearance condition, location, and applicable tolerance	Using nominal thread diameter as the clearance-hole diameter.
Tapped hole	Thread standard, nominal size, pitch, class where required, thread depth, and drill depth if important	Showing only a cylinder without a thread note.
Counterbore	Pilot diameter, counterbore diameter and depth, and fastener head requirement	Forgetting the head depth or the required seating surface.
Countersink	Pilot diameter, included angle, major diameter or screw standard	Using the wrong included angle for the selected fastener.
Dowel/locating hole	Diameter tolerance, position, engagement depth, and datum scheme	Applying an arbitrary loose tolerance to a precision locating feature.

Tolerance allocation

Start with functional requirements: fit, motion, sealing, alignment, strength, and interchangeability. Then select a datum reference frame and assign tolerances to the features that control those requirements. Avoid tight tolerances on non-functional cosmetic dimensions. Use general tolerances for dimensions that do not need individual limits, and explicitly call out critical dimensions and geometric controls.

Size tolerance versus geometric tolerance

Control	What it limits	Example use
Size tolerance	Permissible variation in a feature size	Shaft diameter, plate thickness, hole diameter
Position	Location of a feature relative to datums	Bolt-hole pattern alignment
Flatness	Form of one surface independent of datum reference	Gasket or mounting surface
Perpendicularity	Orientation relative to a datum	Hole axis perpendicular to a mounting face
Parallelism	Orientation relative to a datum	Two mounting faces kept parallel
Runout	Variation during rotation relative to a datum axis	Shaft or pulley surface relative to bearing axis

Fits

A fit describes the relationship between mating feature sizes. A clearance fit leaves space between mating parts; an interference fit creates intentional overlap; a transition fit may produce either small clearance or small interference depending on actual sizes. Select a recognised fit system and tolerance class using the applicable standard and application. Do not assign a fit from intuition alone.

> Tolerance principle: Tighten only what function requires. A drawing with every dimension given a very small tolerance can be difficult and expensive to manufacture and inspect, while still failing to define the true functional relationship.

8. Assembly and inspection

A part must not only be manufacturable; it must also be assemblable and verifiable. Consider tool access, fastener insertion, component orientation, serviceability, and inspection access while designing the CAD model.

Design for assembly checklist

- Reduce unnecessary part count where combining parts does not compromise manufacturing, repair, or material requirements.
- Use self-locating geometry such as shoulders, slots, dowels, and tabs when it improves repeatability.
- Avoid fasteners that require access from an impossible direction or tools that cannot reach the head.
- Provide lead-in chamfers and suitable clearances for insertion where required.
- Prevent incorrect assembly using asymmetric patterns or clear orientation features where practical.
- Consider service access for parts that need replacement or adjustment.

Inspection planning

- Identify critical-to-function characteristics before selecting inspection methods.
- Choose datums that can be physically established on the part and that relate to assembly function.
- Use calipers or micrometers for appropriate size checks, height gauges and surface plates for location checks, bore gauges for internal diameters, and CMM inspection when geometry and tolerance requirements justify it.
- Ensure the measurement method has suitable resolution and uncertainty for the tolerance being inspected.
- Define sample frequency and acceptance criteria in the quality plan, not only on the CAD model.
- Record material, revision, process route, and inspection results for traceability when required.

9. Worked example: sensor mounting bracket

Objective: design a small bracket that supports a sensor and attaches to a machine frame. The example shows how manufacturing choice influences CAD design. The dimensions are illustrative and not a validated product specification.

Initial design brief

Requirement	Illustrative design input	Design implication
Function	Position a compact sensor above a frame datum	Sensor face and mounting interface are functional datums.
Quantity	Prototype batch of 5, possible later production	CNC machining or sheet metal may be practical; volume may change the decision.
Material	Aluminium alloy candidate	Confirm strength, corrosion, finish, and process availability.
Mounting	Two M6 fasteners to frame	Use a standard thread specification and verify edge distances and tool access.
Sensor interface	Two clearance holes or slots	Hole spacing must match sensor datasheet; slots may provide adjustment if permitted.
Environment	Indoor industrial equipment	Consider vibration, contamination, surface finish, and cable clearance.

Option A: CNC-machined bracket

A machined bracket can be designed as a single solid from plate or block stock. Use a base feature that reflects available stock, machine pockets only where weight or clearance requires them, use practical internal radii, and ensure the upright can be held securely. If the bracket is a simple folded shape, compare the one-piece machined design with a sheet-metal version before finalising.

Option B: sheet-metal bracket

For a thin folded bracket, start with a base flange and add an edge flange in the sheet-metal environment. Use a consistent thickness, a supplier-confirmed bend radius, and reliefs where bends meet. Check that the sensor mounting surface remains sufficiently flat after bending. Export the flat pattern only after confirming bend data with the supplier.

CAD-to-manufacturing release sequence

- Create a requirement table from the sensor datasheet and frame interface drawing.
- Choose the process based on quantity, material, stiffness, cost, and available supplier capability.
- Build the CAD model using functional datums and stable references.
- Identify critical hole locations, mounting-face flatness, and any alignment requirements.
- Prepare a drawing that distinguishes machined, formed, and as-supplied surfaces.
- Request supplier feedback on tool access, minimum feature sizes, bend parameters, tolerances, and finish.
- Update the CAD model and drawing after design review; assign a revision and preserve the released native file.
- Inspect the first article against the drawing and record any process adjustments.

Decision point: If the bracket requires a precise, flat sensor datum and a small quantity, CNC machining may be suitable. If it is primarily a thin folded support with modest loads, sheet metal may reduce material use and cycle time. Supplier quotations and engineering checks should decide.

10. Retrieval practice questions

Use these questions to verify that Atlas AI retrieves manufacturing guidance accurately. A strong answer should state when a rule is process-dependent and identify what must be checked with the supplier or governing standard.

Question	Expected answer points
What factors determine the best manufacturing process?	Geometry, material, quantity, properties, tolerances, finish, equipment, tooling, lead time, cost, and secondary operations.
Why should deep narrow pockets be avoided in CNC milling?	They require long tools and can increase deflection, chatter, cycle time, and tool-access problems.
Why do machined internal corners have a radius?	Rotating cutting tools have finite radius; a sharp internal corner requires special relief or another process.
What must be checked before releasing a sheet-metal flat pattern?	Thickness, inside bend radius, bend allowance/K-factor or bend table, reliefs, supplier process, and flat-pattern validity.
Why is build orientation important in additive manufacturing?	It affects strength direction, support need, surface finish, accuracy, and sometimes process success.
What is draft in injection moulding?	A taper on mould-opening faces that helps the part release from the tool. Required draft depends on material, texture, depth, and process.
What is the difference between clearance and interference fits?	Clearance fit leaves space between mating parts; interference fit creates intentional overlap.
Why should tolerances not all be made very tight?	Tight tolerances raise manufacturing and inspection cost and should be reserved for functional requirements.
What should a drawing say about a tapped hole?	Thread standard/designation, nominal size, pitch, and required thread depth; include other details required by the application.
What is the purpose of datum selection?	To define a repeatable reference frame for manufacturing, assembly, and inspection based on function.

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